

Understanding the Impulse Buying Behavior in the Digital Age: Influential Factors in Online Consumer Behavior

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Abstract

Impulse buying is a widely studied consumer behavior influenced by various psychological, social, and marketing factors. This study aims to identify the key drivers of impulse buying within the digital shopping environment. Utilizing the Stimulus-Organism-Response (SOR) framework, this research examines the effects of age, gender, social factors (peer influence, word of mouth), marketing factors (discounts), consumer characteristics (income level, shopping lifestyle), and website-related factors (ease of use, ease of payment) on consumer purchasing decisions. Additionally, the study assesses the mediating role of urgency to buy and explores the cultural and financial influences shaping impulse purchasing behavior in the Filipino market. A quantitative research design was employed, utilizing survey data collected from 381 respondents. Findings indicate a significant relationship between social and website-related factors in driving impulse purchases. Specifically, word of mouth and peer influence enhance website appeal, while ease of use and payment options increase the likelihood of unplanned purchases. Results suggest that the increasing accessibility of online shopping platforms has intensified impulse buying tendencies. This study provides valuable insights for businesses and digital marketers, emphasizing the importance of optimizing website design, integrating customer reviews, and streamlining payment processes to enhance consumer engagement. The findings contribute to the growing body of literature on impulse buying behavior in e-commerce and serve as a foundation for future research on consumer decision-making in digital environments.

CCS Concepts

• **Human-centered computing**; • **User studies, Web-based interaction, and Applied Computing**; • **Electronic Commerce**; • **Online Shopping**;

Keywords

impulse buying, e-commerce, consumer behavior, digital marketing

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1 INTRODUCTION

Impulse buying is a spontaneous purchasing behavior influenced by social, marketing, and website-related factors. Consumers often make unplanned purchases due to emotional triggers, external stimuli, or promotional strategies. Research indicates that 60% of purchases result from impulse buying, highlighting its significant role in consumer decision-making [1].

With the rise of e-commerce and digital marketing, impulse buying has evolved. Consumers are influenced by peer recommendations, personalized ads, and website features that enhance shopping convenience [4]. The COVID-19 pandemic further intensified this behavior, as individuals turned to online shopping for stimulation and social connection [22]. In the Philippine market, impulse buying is also shaped by cultural and social influences, including peer pressure and financial capability [19].

This study examines the key drivers of impulse buying, focusing on the impact of social, marketing, and website-related factors. By understanding these influences, businesses can optimize digital marketing strategies, enhance website usability, and tailor promotions to encourage impulse purchases. These insights contribute to the growing research on consumer behavior in the digital marketplace.

With the rapid evolution of e-commerce and digital marketing, this study examines the key factors driving impulse buying in online shopping. Factors such as discounts, peer influence, and website usability play a significant role in shaping consumer behavior [9]. The study aims to understand how these elements contribute to the shift from impulse buying in physical stores to online platforms, providing insights into optimizing marketing strategies and enhancing consumer engagement. To achieve this, the research seeks to answer the following questions:

1. To what extent do social-related factors, such as word of mouth and peer influence, affect consumer's impulse buying?
2. To what extent do discounts, as a marketing-related factor, affect consumers' impulse buying?



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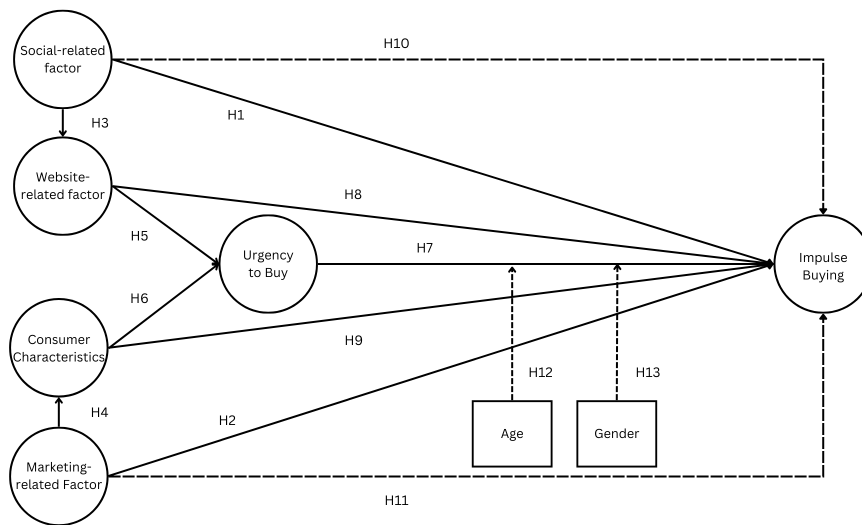


Figure 1: Conceptual Framework

3. To what extent do social-related factors, such as word of mouth and peer influence, affect consumers' perception of websites?
4. To what extent do discounts, as a marketing-related factor, affect consumers' characteristics that lead to impulse buy?
5. To what extent do website-related factors such as ease of use and ease of payment enhance consumers' urgency to buy products online?
6. To what extent do consumer characteristics affect a consumers' urgency to buy online?
7. To what extent does urgency affect a consumer which causes consumers to engage in impulse buying?
8. To what extent do website-related factors affect impulse buying?
9. To what extent do consumer characteristics affect impulse buying?
10. To what extent do website-related factors such as ease of use and ease of payment mediate the effect of social-related factors on impulse buying?
11. To what extent do consumer characteristics like shopping lifestyle mediate the effect of discounts on impulse buying?
12. To what extent does age moderate the effect of urgency to buy on impulse buying?
13. To what extent do different genders moderate the effect of urgency to buy on impulse buying?

The Stimulus-Organism-Response (SOR) framework explains how external stimuli, such as marketing strategies, website design, and promotions, influence consumer behavior through internal cognitive and emotional processing, leading to impulse buying [6] [12]. Originally derived from the stimulus-response model, the SOR theory integrates stimuli, organism responses, and behavioral outcomes to provide a deeper understanding of purchasing decisions [13] [16].

Using the Stimulus-Organism-Response theory, researchers developed a conceptual framework incorporating social-related factors, marketing-related factors, website-related factors, consumer characteristics, and urgency to buy. These elements act as stimuli, influencing the organism (consumer) and generating responses that lead to impulse buying behavior in online shopping.

2 THE PRESENT STUDY

2.1 Social-related Factor and Impulse Buying

Consumer purchasing decisions are often influenced by social factors, including peer opinions, recommendations, and experiences shared through electronic word of mouth (eWOM) and social media [10]. Social influence affects consumer behavior, shaping beliefs, attitudes, and purchasing decisions [1] [10]. Peer influence plays a significant role, as individuals tend to align their choices with their social circles [12]. Additionally, word-of-mouth marketing remains a powerful tool in consumer decision-making, as personal recommendations from trusted sources enhance product credibility and encourage impulse buying [13]. Research highlights, that social interactions, particularly through peers and digital platforms, drive impulse buying behavior by reinforcing trust and perceived product value [15].

H1: Social-related factors have a significant effect on impulse buying

2.2 Marketing-related Factor and Impulse Buying

Impulse buying is an unplanned purchase driven by spontaneous decisions. Marketers used factors like discounts, promotions, and product appeal to trigger these impulses [14]. Studies show that discounts significantly influence impulse buying, though some findings suggest otherwise, highlighting a research gap [17] [23]. Promotions such as price cuts, coupons, and flash sales enhance consumer interest, often leading to impulsive purchases [5] [16]. With

growing online shopping trends, understanding these marketing influences remains crucial [7].

H2: Marketing-related factors significantly affect impulse buying

2.3 Social-related Factors and Website-related Factors

Social-related factors significantly influence website-related factors by shaping consumer perceptions and interactions in digital environments. Li (2024) emphasized that product recommendations from trusted video creators foster emotional connections, enhancing consumer trust and purchase intent, whereas recommendations from less favorable sources may lead to product rejection [13]. Engaging advertisements and new media marketing strategies subtly reshape consumer preferences, reinforcing purchase motivation [20]. As website experiences are influenced by social interactions, including peer influence, social presence, and cultural context, which affect perceived usefulness, ease of use, and overall engagement [10], these findings underscore the importance of integrating social elements into website design to optimize user experiences and strengthen consumer trust [24].

H3: Social-related factors have a significant effect on website-related factors

2.4 Marketing-related Factor and Consumer Characteristics

Marketing-related factors influence consumer characteristics by shaping perceptions of discounts and promotions. Emotion-driven consumers see discounts as enhancing satisfaction, while concerns over quality and authenticity can discourage purchases [14] [20]. As for income levels affect discount tolerance, but product type and necessity also play key roles [19].

H4: Marketing-related factors have a significant effect on consumer characteristics

2.5 Website-related Factors and Urgency to Buy

The growing prevalence of e-commerce has underscored the significance of website-related factors in driving purchase urgency. Research shows that website design elements, including high-quality visuals, structured layouts, and scarcity-driven mechanisms such as countdown timers and limited-stock notifications, significantly influence consumer decision-making [9]. A well-designed website creates trust and enhances the perceived value of urgency cues, enhancing impulse buying behaviors through the fear of missing out [13] [22]. As digital commerce continues to evolve, the strategic integration of these elements remains essential in shaping consumer purchasing patterns.

H5: Website-related factors have a significant effect on urgency to buy

2.6 Consumer Characteristics and Urgency to Buy

Impulse buying urgency is influenced by consumer characteristics such as impulsivity, shopping lifestyle, and fashion involvement. Highly impulsive individuals are more susceptible to purchasing

triggers, while trend-driven consumers respond quickly to time-limited offers and product scarcity [9] [17]. The need to align with evolving fashion trends further amplifies this urgency [20] [25]. While strategic urgency tactics can drive impulse purchases, excessive pressure may diminish consumer satisfaction and long-term engagement [17].

Hypothesis 6: Consumer Characteristics has a significant effect on Urgency to Buy

2.7 Urgency to Buy and Impulse Buying

Urgency, whether real or artificial, drives impulse buying by triggering emotions like fear of missing out and anticipated regret [6] [9]. Time-limited offers, scarcity cues, and price fluctuations heighten this effect, pushing consumers toward quick, emotion-driven purchases [5]. Retailers strategically use these tactics, particularly in fast fashion and digital commerce, to accelerate decision-making and increase sales [9].

Hypothesis 7: Urgency to Buy has a significant effect on Impulse Buying

2.8 Website-related Factors and Impulse Buying

The evolution of digital markets has enhanced accessibility, influencing impulse buying through website-related factors such as ease of use, accessibility, and seamless payment [24]. Intuitive navigation and frictionless transactions reduce cognitive effort, increasing unplanned purchases. However, some studies suggest that excessive ease may diminish excitement, reducing impulsive tendencies [17]. While individual consumer traits influence responses, website design remains a key driver of impulse buying behavior.

Hypothesis 8: Website-related Factors has a significant effect on Impulse Buying

2.9 Consumer Characteristics and Impulse Buying

Shopping lifestyle strongly influences impulse buying, with frequent shoppers and those who see shopping as entertainment being more prone to unplanned purchases [20]. Online browsing, tailored ads, and promotions further drive impulsive decisions [20]. Hedonic shopping motivation also plays a key role, as consumers seek emotional gratification through spontaneous buys [20]. While higher income increases purchasing power [19], financial literacy and budgeting habits help moderate impulse buying [2].

Hypothesis 9: Consumer Characteristics has a significant effect on Impulse Buying

2.10 Website-related Factor as Mediator of Social-related Factor and Impulse Buying

Website-related factors enhance impulse buying by improving usability, convenience, and emotional engagement. A well-designed website with intuitive navigation, fast-loading pages, and seamless e-payment systems encourages spontaneous purchases [23] [24]. With positive user experiences drive offline word-of-mouth, it enhances impulse buying behavior [24]. Aligned with the SOR framework, website quality strengthens social influence, leading to unplanned purchases [10].

Hypothesis 10: Website-related Factor mediates the relationship of Social-related Factor and Impulse Buying

2.11 Consumer Characteristics as Mediator of Marketing-related Factor and Impulse Buying

Consumer shopping lifestyle mediates the impact of discounts on impulse buying, as hedonic shoppers respond strongly to promotional stimuli, triggering urgency and unplanned purchases [3]. Perceived savings influence decision-making, particularly among leisure-driven shoppers [26]. This highlights shopping lifestyle's role in shaping impulse buying and optimizing marketing strategies [15].

Hypothesis 11: Consumer Characteristics mediates the relationship of Marketing-related Factor and Impulse Buying

2.12 Age as moderator between Urgency to Buy and Impulse Buying

Younger consumers (18–30) are more prone to impulse buying due to social media influence and the appeal of instant gratification [13] [26]. Platforms like Instagram and TikTok encourage spontaneous purchases through visually engaging content [25]. In contrast, individuals aged 30–45 exhibit greater restraint, prioritizing financial stability, though promotional cues can still trigger impulse buys [21]. However, promotional cues such as discounts can still influence impulse purchases in this group, though with a more calculated approach.

Hypothesis 12: Age moderates the relationship between Urgency to Buy and Impulse Buying

2.13 Gender as moderator between Urgency to Buy and impulse Buying

The evolving concept of masculinity has influenced male shopping behaviors, leading to increased engagement in traditionally feminine consumption categories such as fashion and skincare [9] [18]. While men typically prioritize efficiency and urgency to minimize the perception of excessive focus on appearance, women tend to engage in more deliberate shopping behaviors, emphasizing comparison and informed decision-making [9]. This shift reflects changing gender norms, with younger men adopting shopping habits once associated with female consumers, driven in part by the growing emphasis on urgency in purchasing decisions.

Hypothesis 13: Gender moderates the relationship between Urgency to Buy and Impulse Buying

3 METHODOLOGY

3.1 Design and Instrument

This study utilized a quantitative research design to examine the factors influencing impulsive buying behavior. By applying a correlational approach, the research aimed to determine the relationships between demographic variables, social-related factors, and website-related factors in driving impulsive purchases. This method provided valuable insights into consumer behavior, contributing to a deeper understanding of online shopping behavior in Cabuyao, Laguna.

Researchers employed a structured, self-administered questionnaire distributed both online and in person. The survey was divided into multiple sections, covering demographic characteristics and key behavioral factors. It included seven variables, measured using a five-point Likert scale using google forms for digital distribution, ensuring convenience, accessibility, and broad participant reach.

3.2 Sampling Technique and Sample Size

Purposive sampling was utilized in this study to ensure the selection of participants who met specific criteria relevant to the research objectives. This non-probability sampling method allowed researchers to target individuals whose characteristics aligned with the study's focus on impulsive buying behavior. Participants were selected based on age groups—21–30 (Gen Z) and 31–40 (Millennials)—and their experience with online shopping and website usability.

A total of 381 respondents from Cabuyao, Laguna were surveyed, exceeding the minimum requirement of 59 based on Cohen's Table, with an effect size of 0.25, an alpha level of 5%, and a statistical power of 80% [8]. Data collection was conducted through Google Forms and QR codes to ensure accessibility. Confidentiality was strictly maintained in compliance with the Data Privacy Act of 2012, enhancing the reliability and ethical integrity of the study.

3.3 Statistical Treatment

The study utilized Partial Least Squares Structural Equation Modeling (PLS-SEM) to analyze structural relationships among variables, particularly in cases with small sample sizes and non-normal data distributions. WarpPLS 8.0 software was used to validate and assess the reliability of the measurement model. The evaluation included Composite Reliability (CR) and Cronbach's Alpha (CA), ensuring construct reliability, as well as convergent and discriminant validity tests to confirm the clarity and accuracy of latent variables.

Convergent validity was assessed through the Average Variance Extracted (AVE), requiring values above 0.5 for acceptance. Discriminant validity was evaluated using the Fornell-Larcker criterion, ensuring that each variable was distinct. Additionally, the Variance Inflation Factor (VIF) was examined to detect multicollinearity, with an acceptable threshold of 3.3 to prevent common method bias. R-squared values greater than 0.5 indicated strong explanatory power, confirming the model's effectiveness in predicting outcomes.

Discriminant validity is assessed according to a specific criterion to validate its acceptability. The average variance extracted (AVE) should have a square root greater than any correlations between the latent variables. Thus, the values along the diagonal must exceed the values in the rows to their left or right.

The Variance Inflation Factor (VIF) assesses multicollinearity in structural models, where high correlations between independent variables can affect model stability. A VIF below 3.3 indicates no collinearity issues, while values under 10 suggest an acceptable range [11]. The R^2 values—0.754 for website-related factors, 0.510 for consumer characteristics, 0.406 for urgency to buy, and 0.804 for impulse buying—demonstrate strong predictive accuracy, exceeding Cohen's (1988) threshold for large effects [8]. Moreover, the Stone-Geisser test (Q^2) confirms the model's predictive relevance, as all values are above zero.

Table 1: Convergent Validity and Reliability Measures

Construct	Ave Variance Extract (AVE)	Composite Reliability	Cronbach Alpha
Social-related Factors	0.745	0.936	0.914
Marketing-related Factor	0.691	0.947	0.936
Website-related Factors	0.811	0.945	0.922
Consumer Characteristics	0.547	0.857	0.791
Urgency to Buy	0.717	0.927	0.901
Impulse Buying	0.675	0.943	0.931

Table 2: Discriminant Validity using Fornell and Larcker Criterion.

Constructs	SRF	MRF	WRF	CC	UTB	IB
Social-related Factors	0.836					
Marketing-related Factor	0.861	0.900				
Website-related Factor	0.721	0.695	0.740			
Consumer Characteristics	0.795	0.800	0.663	0.831		
Urgency to Buy	0.519	0.477	0.632	0.551	0.847	
Impulse Buying	0.392	0.378	0.540	0.429	0.821	0.822

Table 3: Discriminant validity using Fornell and Larcker Criterion.

Constructs	Full Collinearity	R ²	Q ²
Social-related Factors	5.105		
Marketing-related Factor	3.655		
Website-related Factors	4.695	0.754	0.742
Consumer Characteristics	2.739	0.510	0.493
Urgency to Buy	3.852	0.406	0.406
Impulse Buying	3.219	0.804	0.685

4 RESULTS AND DISCUSSION

The study confirms significant relationships between social-related factors, marketing-related factors, website usability, urgency to buy, and impulse buying. However, hypotheses H1, H2, H8, H12, and H13 are rejected. The findings highlight the role of website-related factors and consumer characteristics in influencing urgency and impulse buying. For significance, p-values must be below 0.01, and effect sizes indicate varying strengths of relationships. The mediation model results further validate indirect effects on impulse buying.

A final model (2) that removes factors with statistically insignificant results is the result of social-related factor, marketing-related factor, website-related factor, consumer characteristics, urgency to buy, and impulse buying.

5 CONCLUSION AND RECOMMENDATIONS

This study assessed the impact of social, marketing, and website-related factors, consumer characteristics, and urgency to buy on impulse buying in online shopping. Findings indicate that marketing and social factors do not directly drive impulse buying but influence consumer characteristics and website usability, respectively. Website-related factors significantly affect urgency to buy,

which directly predicts impulse buying. Consumer characteristics also play an important role which emphasizes the importance of individual shopping behaviors. However, age and gender were not found to moderate the urgency-impulse buying relationship.

These findings highlight the role of website usability and urgency-driven strategies in enhancing impulse purchases. Social factors attract consumers to platforms, but purchase decisions are largely shaped by website experience and time-sensitive promotions. Based on the findings, it is recommended that businesses should focus on seamless website navigation, multiple payment options, and real-time stock updates to encourage impulse purchases. Marketers should utilize personalized promotions, such as flash sales and limited-time offers, while ensuring strategic product placements and clear discount visibility. Additionally, website developers should prioritize usability enhancements to create an engaging and frictionless shopping experience.

Furthermore, future researchers should explore broader demographic and cultural contexts, as well as additional marketing factors like social media advertising and personalized recommendations. Investigating psychological factors such as self-control and emotional responses could further deepen understanding of impulse buying behaviors. By optimizing e-commerce strategies based on

Table 4: Evaluation of the Structural Model

Path Coefficients	β	SE	p-value	f^2	Interpretation	Effect size
Direct Correlation						
H1. SRF → IB	0.047	0.051	0.176	0.019	Accept	
H2. MRF → IB	0.064	0.051	0.104	0.029	Accept	
H3. SRF → WRF	0.868	0.045	<0.001	0.754	Reject	Large
H4. MRF → CC	0.714	0.046	<0.001	0.510	Reject	Large
H5. WRF → UTB	0.099	0.051	0.026	0.050	Reject	Small
H6. CC → UTB	0.562	0.047	<0.001	0.356	Reject	Large
H7. UTB → IB	0.804	0.046	<0.001	0.660	Reject	Large
H8. WRF → IB	0.031	0.051	0.269	0.013	Accept	
H9. CC → IB	0.109	0.050	0.016	0.061	Reject	Small
Mediating / Indirect Effects						
H10. SRF → WRF → IB	0.080	0.036	0.014	0.032	Reject	Small
H11. MRF → CC → IB	0.452	0.034	<0.001	0.252	Reject	Medium
H12. AGE* → UTB → IB	0.044	0.051	0.192	0.013	Accept	
H13. GENDER* → UTB → IB	0.025	0.051	0.311	0.010	Accept	

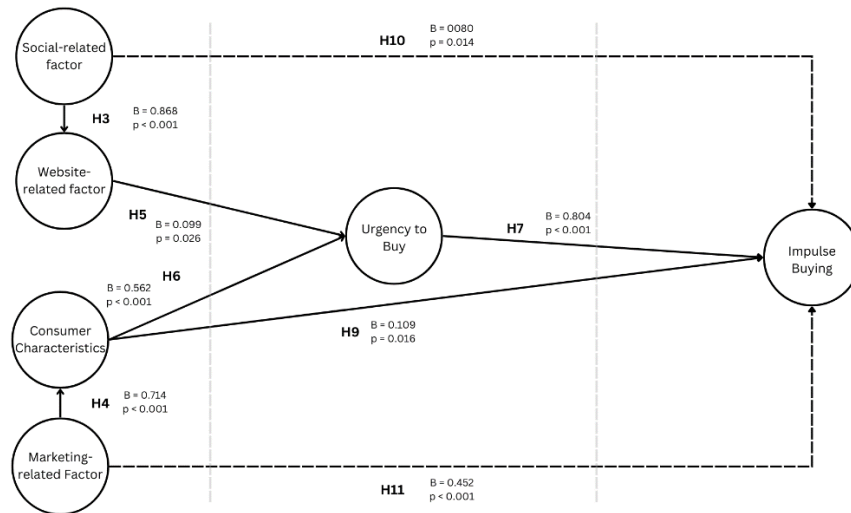


Figure 2: Final Model

these insights, businesses can improve customer engagement, enhance satisfaction, and drive higher sales.

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References

- [1] Aprilia, L., & Wibowo, L. A. (2016). Pengaruh faktor individu dan sosial terhadap (survei pada pengguna aplikasi mobile messaging di Kota Bandung) keputusan menggunakan. *Journal of Business Management Education (JBME)*, 1(1), 205–215. <https://doi.org/10.17509/jbme.v1i1.2289>
- [2] Azul, R. D., Cadano, K. D. Q., Guiao, R. J. M., Maranga, N., Salonga, C. H., & Usita, M. N. (2023). The impulse buying behavior and financial well-being profile of young professionals in the Philippines. *International Journal of Scientific and Research Publications*, 13(3), 308–314. <https://doi.org/10.29322/IJSRP.13.03.2023.p13537>
- [3] Beatty, S. E., & Ferrell, M. E. (1998). Impulse buying: Modeling its precursors. *Journal of Retailing*, 74(2), 169–191. [https://doi.org/10.1016/S0022-4359\(99\)80092-X](https://doi.org/10.1016/S0022-4359(99)80092-X)
- [4] Cavazos-Arroyo, J., & Máñez-Guaderrama, A. (2022). Antecedents of online impulse buying: An analysis of gender and centennials' and millennials' perspectives. <https://doi.org/10.3390/jtaer17010007>
- [5] Chan, K. K. (1996). Chinese viewers' perception of informative and emotional advertising. *International Journal of Advertising*, 15(2), 152–166. <https://doi.org/10.1080/02650487.1996.11104644>
- [6] Chandrasekhar, K., Das, S., & Gupta, N. (2024). Impulse buying and retailer loyalty programs: Assess the influence of loyalty programs, rewards, and customer incentives on impulse buying. *International Journal of Research - Granthaalayah*, 12(8). <https://doi.org/10.29121/granthaalayah.v12i8.2024.5748>
- [7] Chen, X., Parsad, C., Prashar, S., Vijay, S., & Kumar, M. (2021). Do promotion and prevention focus influence impulse buying: The role of mood regulation, shopping values, and impulse buying tendency. *Journal of Retailing and Consumer Services*, 61, 102567. <https://doi.org/10.1016/j.jretconser.2021.102567>

- [8] Cohen, J. (1988). *Statistical power analysis for behavioral sciences* (2nd ed.). Lawrence Erlbaum. <https://doi.org/10.4324/9780203771587>
- [9] Gupta, S. (2013). The psychological effects of perceived scarcity on consumers' buying behavior. *Dissertations, Theses, and Student Research from the College of Business*. <https://digitalcommons.unl.edu/businessdiss/41>
- [10] Gunawan, C., Rahmania, L., & Hendrika, I. (2023). The influence of social influence and peer influence on intention to purchase in e-commerce. *Review of Management and Entrepreneurship*, 7(1), 61-84. <https://doi.org/10.37715/rme.v7i1.3683>
- [11] Hair, J. F., Black, W. C., Babin, B. J., Anderson, R. E., & Tatham, R. L. (2006). *Multivariate data analysis* (6th ed.). Pearson.
- [12] Khare, A., & Pandey, S. (2017). Role of green self-identity and peer influence in fostering trust towards organic food retailers. *International Journal of Retail & Distribution Management*, 45(9), 969–990. <https://doi.org/10.1108/IJRDM-07-2016-0109>
- [13] Le, N., Mai, N. T. B., Nguyen, N. Y. T., & Vu, H. N. T. (2024). Multi-sensory marketing and impulsive buying behavior: The role of impulsive buying traits. *Emerging Science Journal*, 8(5), 1764–1782. <https://doi.org/10.28991/esj-2024-08-05-05>
- [14] Lee, L., & Tsai, C. I. (2014). How price promotions influence postpurchase consumption experience over time. *Journal of Consumer Research*, 40, 943–959. <https://doi.org/10.1086/673441>
- [15] Liu, Z., & Lu, Z. (2017). Research on influence of shopping app's characteristic on consumer's impulse buying. *Modern Economy*, 8(12), 1509-1518. <https://doi.org/10.4236/me.2017.812099>
- [16] Madhavaram, S. R., & Laverie, D. A. (2004). Exploring impulse purchasing on the internet. In *NA-Advances in Consumer Research Volume 31* (pp. 59-66). Association for Consumer Research.
- [17] Ngo, T. T. A., Nguyen, H. L. T., Nguyen, H. P., Mai, H. T. A., Mai, T. H. T., & Hoang, P. L. (2024). A comprehensive study on factors influencing online impulse buying behavior: Evidence from Shopee video platform. *Heliyon*, 10(15), e35743. <https://doi.org/10.1016/j.heliyon.2024.e35743>
- [18] Ostberg, J. (2009, June). Conservative rebels: Identity struggles by resurrecting an imagined past. *Consumer Culture Theory Conference*, Ann Arbor, MI.
- [19] Sangalang, R., Siochi, A., & Fernandez, M. (2017). The impact of income level on impulse buying behavior: An analysis of purchasing power and consumer decision-making. [https://doi.org/\[DOI\]](https://doi.org/[DOI])
- [20] Sari, I., & Kusumawati, A. (2021). The influence of shopping lifestyle, fashion involvement, and hedonic shopping on e-commerce with the moderating role of impulse buying. *Global Business and Management Research: An International Journal*, 16(2s), 59-70. <https://gbmrjournal.com/pdf/v16n2s/V16N2s-59.pdf>
- [21] Sohn, Y. S., & Ko, M. T. (2021). The impact of planned vs. unplanned purchases on subsequent purchase decision making in sequential buying situations. *Journal of Retailing and Consumer Services*, 61, 102020. <https://doi.org/10.1016/j.jretconser.2021.102020>
- [22] Sundström, A., Bergström, K., & Hagberg, J. (2019). Clicking the boredom away: Exploring impulse fashion behavior online. *Journal of Retailing and Consumer Services*, 49, 133–140. <https://doi.org/10.1016/j.jretconser.2019.01.010>
- [23] Utami, W. N., & Juanda, T. A. (2022). The effect of price discount and electronic word of mouth on impulse buying at marketplace Shopee. *Journal of Business and Management INABA*, 1(2), 136–152. <https://doi.org/10.56956/jbmi.v1i02.122>
- [24] Widagdo, B., & Roz, K. (2021). Hedonic shopping motivation and impulse buying: The effect of website quality on customer satisfaction. *Journal of Asian Finance, Economics, and Business*, 8(1), 395. <https://doi.org/10.13106/jafeb.2021.vol8.no1.395>
- [25] Xu, Y., Dzever, S., & Zhao, G. (2023). Measuring the effects of pressure on consumer impulse buying intention in online group buying. *International Journal of Enterprise Information Systems*, 19(1), 1–23. <https://doi.org/10.4018/IJEIS.326549>
- [26] Youn, S., & Faber, R. (2000). Impulse buying: Its relation to personality traits and cues. *Advances in Consumer Research*, 27, 179-185.